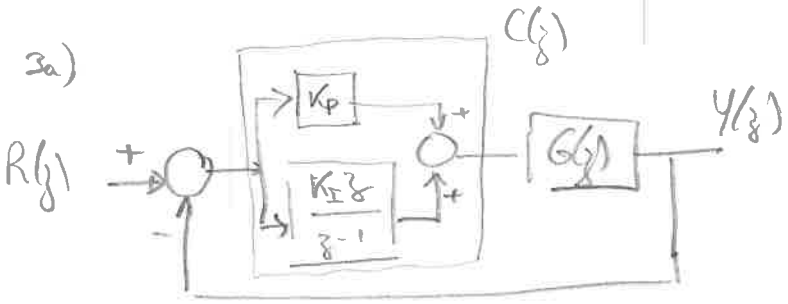
Ex II — LIE421 – Eléments de conception – Examen partiel du 03/11/2020 (ST)

1) $G(z) = \dfrac{z-1}{z^2-3z+2} = \dfrac{z-1}{(z-1)(z-2)} = \dfrac{1}{z-2}$

2) $p_1 = 1 \qquad p_2 = 2 \qquad |p_i| \geq 1 \rightarrow$ instable EBSB

3a)



$$C(z) = \frac{(K_P + K_I)z - K_P}{z-1}$$

(3b) Eq. caract. : $1 + G(z)C(z) = 0$.

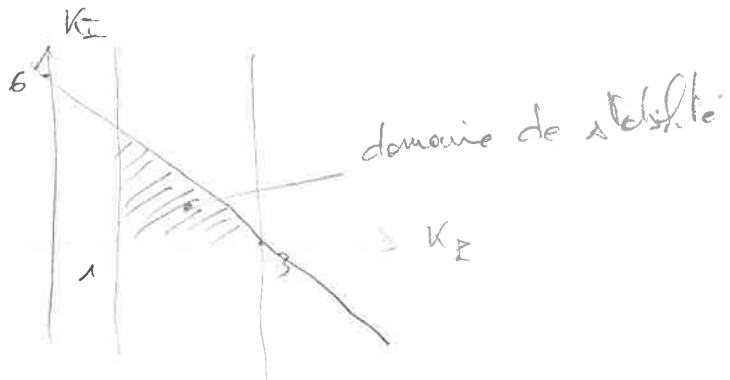
$$\Leftrightarrow (z-1)(z-2) + (K_P + K_I)z - K_P = 0$$

$$\Leftrightarrow z^2 + (K_P + K_I - 3)z + 2 - K_P = 0$$

3c) i) $1 + K_P + K_I - 3 + 2 - K_P > 0 \Leftrightarrow K_I > 0$

ii) $1 - K_P - K_I + 3 + 2 - K_P > 0 \Leftrightarrow 6 > 2K_P + K_I$

iii) $-1 < 2 - K_P < 1 \Leftrightarrow -1 < K_P - 2 < 1 \Leftrightarrow 1 < K_P < 3$



$$H(p) = \frac{K}{(p+a)^2}$$

1) $$A(p) = \frac{H(p)}{p} = \frac{K}{p(p+a)^2} = K\left(\frac{\alpha}{p} + \frac{\beta}{p+a} + \frac{\gamma}{(p+a)^2}\right)$$

$\gamma = -1/a \quad \alpha = 1/a^2 \quad \alpha+\beta = 0 \quad \beta = -1/a^2$

$$A(p) = \frac{K/a^2}{p} - \frac{K/a^2}{p+a} - \frac{K/a}{(p+a)^2}$$

$$a(t) = \frac{K}{a^2}\left(1 - e^{-ta} - ate^{-ta}\right)$$

$$a_k = \frac{K}{a^2}\left(1 - e^{-T_e a k} - k a T_e e^{-T_e a k}\right) \qquad \underline{D := e^{-T_e a}}$$

$$A(z) = \frac{K}{a^2}\left(\frac{z}{z-1} - \frac{z}{z-D} - \frac{a T_e D z}{(z-D)^2}\right)$$

$$G(z) = \frac{z-1}{z} A(z) = \frac{K}{a^2}\left(1 - \frac{z-1}{z-D} - \frac{a T_e D (z-1)}{(z-D)^2}\right)$$

$$= \frac{K}{a^2}\left(\frac{z^2 - 2Dz + D^2 - z^2 + (1+D)z - D - aT_eD(z-1)}{(z-D)^2}\right)$$

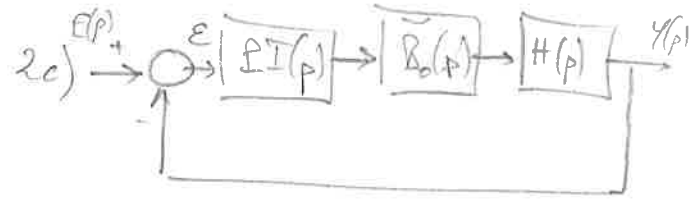
$$= \frac{K}{a^2}\,\frac{(1-D-aT_eD)z + (D^2 - D + aT_eD)}{(z-D)^2}$$

$$b_1 = \frac{K}{a^2}\left(1 - D(1+aT_e)\right) \qquad b_0 = \frac{K}{a^2} D\left(D - 1 + aT_e\right)$$

2a) $$\tilde{B}_0(p) = e^{-\frac{T_e}{2}p}$$

2b) $$PI(p) = K_p\left(1 + \frac{1}{T_i p}\right)$$

2c)



2d) $$\varepsilon(p) = \frac{1}{1 + PI(p)\tilde{B}_0(p)H(p)} E(p) = \frac{T_i p (p+a)^2}{T_i p (p+a)^2 + K_p(1+T_i p) K e^{-\frac{T_e}{2}p}} E(p)$$

pour $E(p) = \frac{E_0}{p}$ $\quad \varepsilon_s = \lim_{p\to 0} p\,\varepsilon(p) = \frac{0}{K_p K} \cdot E_0 = 0 \quad \underline{\forall E_0 \in \mathbb{R}}$

2f) Euler avant : $p \leftarrow \frac{z-1}{T_e}$

2e) $C(z) = PI(p)\Big|_{p=\frac{z-1}{T_e}} = K_p\left(1 + \frac{T_e}{T_i}\frac{1}{z-1}\right) = K_p\left(\frac{z + \frac{T_e}{T_i} - 1}{z-1}\right)$

$K_p = 3{,}1623$, $T_i = 20$

$$C(z) = 3{,}1623 \cdot \frac{z - 0{,}975}{z-1} = K_p \cdot \frac{z-\alpha}{z-1}$$

2g) Eq. caract. de la bf : $1 + C(z)G(z) = 0$

$\Leftrightarrow (z-1)(z-D)^2 + K_p(z-\alpha)(b_1 z + b_0) = 0$

$\Leftrightarrow (z-1)(z^2 - 2Dz + D^2) + K_p(b_1 z^2 + (b_0 - \alpha b_1)z - \alpha b_0) = 0$

$\Leftrightarrow z^3 - (1+2D)z^2 + (D^2+2D)z - D^2 + K_p(b_1 z^2 + (b_0 - b_1\alpha)z - \alpha b_0) = 0$

$\Leftrightarrow z^3 + (K_p b_1 - 1 - 2D)z^2 + (D(D+2) + K_p(b_0 - b_1\alpha))z - D^2 - K_p \alpha b_0 = 0$.

$c_3 = 1$

$c_2 = K_p b_1 - 1 - 2D \;\#\; -2{,}9286$

$c_1 = D(D+2) + K_p(b_0 - b_1\alpha) \;\#\; 2{,}5418$

$c_0 = -D^2 - K_p \alpha b_0 \;\#\; -0{,}81144$

i) $c_3 + c_2 + c_1 + c_0 \;\#\; 0{,}0019465 > 0 \longrightarrow$ OK

ii) $c_3 - c_2 + c_1 - c_0 \;\#\; 7{,}0818 > 0$

iii) $-c_3 < c_0 < c_3 \longrightarrow -1 < -0{,}74616 < 1 \longrightarrow$ OK

iv) $c_0^2 - c_3^2 < c_0 c_2 - c_1 c_3 \longrightarrow y - x \;\#\; 0{,}013892 > 0 \longrightarrow$ OK

$x < y$

3a) $\tilde{G}(w) = G(z)\Big|_{z=\frac{1+w}{1-w}}$

$$= \frac{b_1\left(\frac{1+w}{1-w}\right) + b_0}{\left(\frac{1+w}{1-w} - D\right)^2} = \frac{b_1(1-w^2) + b_0(1-w)^2}{\left((1+D)w + 1 - D\right)^2}$$

$$= \frac{(b_0 - b_1)w^2 - 2b_0 w + b_0 + b_1}{\left((1+D)w + 1 - D\right)^2}$$

3b) CdC mélangé

$\tilde{\omega} = \tan\frac{\omega T_e}{2}$. $\forall \omega \in \left[0\ \frac{\pi}{T_e}\right]$ bijectif.

$+ \lim_{z\to 1}(z-1)E(z) = \lim_{w\to 0}\frac{2w}{1-w}E(z)\Big|_{z=\frac{1+w}{1-w}}$

3c) $PI(\tilde{\omega}) = \tilde{K}_p\left(1 + \frac{1}{\tilde{T}_i \tilde{\omega}}\right)$

$\tilde{K}_p \simeq 3,5481 \qquad \tilde{T}_i \simeq 76,923$

3d) $\tilde{C}(z) = PI(w)\Big|_{w=\frac{z-1}{z+1}} = \tilde{K}_p \frac{z-1+1/\tilde{T}_i(z+1)}{z-1} = \tilde{K}_p \frac{(1+\frac{1}{\tilde{T}_i})z - (1-\frac{1}{\tilde{T}_i})}{z-1}$

$$= \tilde{K}_p\left(1 + \frac{1}{\tilde{T}_i}\right) \frac{z - \frac{\tilde{T}_i - 1}{\tilde{T}_i + 1}}{z-1} = \hat{K}_p \frac{z-\hat{\alpha}}{z-1}$$

$\hat{K}_p = 3,5943$

$\hat{\alpha} = 0,97433$

vs

$K_p = 3,1623$

$\alpha = 0,975$

3e) Pas d'approx. du modèle, juste dégt de variable bijectif.

Ex IV

1) $Y(z) = \frac{BT}{AR+BS} E(z) + \frac{BR}{AR+BS} W(z)$

2) a) $A_m = (z - p_{d_1})(z - p_{d_2})$ $\quad p_{d_j} = e^{p_j T_e}$ $\quad j = 1, 2$

$= z^2 - (e^{-0,1} + e^{-0,2}) z + e^{-0,3} = z^2 + c_1 z + c_0$.

b) $q - \mu \geq m - m = 1$: $B(z) = z + 2$, zéro instable $\Rightarrow$ pas de compensation envisageable.

c) $\begin{cases} BT = B_d \\ AR + BS = A_m \end{cases}$ $\quad B_d = B \cdot \tilde{B}$ où $\mu = \deg B_d \leq q - 1 = 1$

puisque $\deg B = 1$

$\Rightarrow \deg \tilde{B}_d = 0$

$\Rightarrow \tilde{B}_d(z) = K$

d) pour $E(z) = 0$ et $W(z) = \frac{W_0 z}{z-1}$

on veut $\lim_{z \to 1} (z-1) Y(z) = 0 \Leftarrow \lim_{z \to 1} (z-1) \frac{BR}{AR+BS} \frac{W_0 z}{z-1} = 0 \Leftrightarrow \boxed{R(z) = (z-1)\tilde{R}(z)}$

e) $\tilde{A}\tilde{R} + BS = A_m$

$\tilde{A}(z) = (z-1)A(z) \rightarrow \tilde{n} = n+1$

$\begin{cases} \tilde{n} + \tilde{p} = q \\ \tilde{p} + \sigma + 1 = q \\ p \geq \sigma \end{cases} \rightarrow \sigma = \tilde{n} - 1 = n + 1 - 1 = 2$

$\rightarrow \tilde{p} + 1 \geq n \Leftrightarrow q - \tilde{n} + 1 \geq n$

$q \geq n + \tilde{n} - 1$

$\Rightarrow A_0(z) = (z - p_0)^{k_2}$

3a) $\begin{cases} \tilde{A}\tilde{R} + BS = A_m A_0 \\ BT = B_d A_0 \end{cases}$ $\quad \begin{cases} \tilde{n} + \tilde{p} = q + k \\ \tilde{p} + \sigma + 1 = q + k \\ p \geq \sigma \end{cases}$ $\quad \begin{cases} \sigma = \tilde{n} - 1 = n = 2 \\ \tilde{p} + 1 \geq n \Leftrightarrow q + k - \tilde{n} + 1 \geq n \end{cases}$

$\boxed{k \geq 2n - q = 2}$

Poser $\Pi_d(z) = A_m(z) A_0(z)$

$= (z^2 + c_1 z + c_0)(z^2 - 2p_0 z + p_0^2) = z^4 + \underbrace{(c_1 - 2p_0)}_{\tilde{c}_3} z^3 + \underbrace{(p_0^2 - 2c_1 p_0 + c_0)}_{\tilde{c}_2} z^2$

$+ \underbrace{(c_1 p_0^2 - 2c_0 p_0)}_{\tilde{c}_1} z + \underbrace{c_0 p_0^2}_{\tilde{c}_0}$

Choix de p_0 : $\max\{|p_{d_j}|\} \ll |p_0| < 1$ $\quad$ ex: $\boxed{p_0 = e^{T_e q_0}}$

où $q_0 < -2$ et $q_0 > -\frac{\pi}{T_e} \simeq -3,415$, ex: $q_0 = -20$.

$\tilde{A}(z) = (z-1)(z+1)(z-2) = (z^2 - 1)(z-2) = z^3 - 2z^2 - z + 2$

$= z^3 + \tilde{a}_2 z^2 + \tilde{a}_1 z + \tilde{a}_0$

$\tilde{A}\tilde{R} + BS = A_m A_0$

$k = 2$

$\tilde{p} = 1$ $\quad \tilde{R}(z) = z + \tilde{r}_0$

$\sigma = 2$ $\quad S(z) = s_2 z^2 + s_1 z + s_0$

3b) d'où le syst. fn. à résoudre.

(ST) - TV : 3/2

$$\begin{bmatrix} 1 & b_1 & 0 \\ \tilde{a}_2 & b_0 & b_1 & 0 \\ \tilde{a}_1 & 0 & b_0 & b_1 \\ \tilde{a}_0 & 0 & 0 & b_0 \end{bmatrix} \begin{bmatrix} \tilde{\pi}_0 \\ \lambda_2 \\ \lambda_1 \\ \lambda_0 \end{bmatrix} = \begin{bmatrix} \tilde{c}_3 - \tilde{a}_2 \\ \tilde{c}_2 - \tilde{a}_1 \\ \tilde{c}_1 - \tilde{a}_0 \\ \tilde{c}_0 \end{bmatrix}$$

3c) $BT = B_d A_0 \iff BT = B.K A_0$

$$\iff T(z) = K A_0(z) = K(z - p_0)^2$$

On veut $\lim_{z \to 1} (z-1) Y(z) = t_0$ pour $E(z) = \frac{t_0 z}{z-1}$

$$\lim_{z \to 1} (z-1) \frac{B_d}{A_m} \frac{t_0 z}{z-1} = t_0 \iff \lim_{z \to 1} \frac{B_d(1)}{A_m(1)} = 1$$

$$\frac{B_d(1)}{A_m(1)} = \left. \frac{K(z+2)}{(z-p_{d_1})(z-p_{d_2})} \right|_{z \to 1} = \frac{3K}{(1-p_{d_1})(1-p_{d_2})} = 1 \iff \boxed{K = \frac{(1-p_{d_1})(1-p_{d_2})}{3}}$$